

AYUSH RAJ

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Summary

B.Tech student at IIIT Bhubaneswar with hands on experience in Machine Learning, NLP, Generative AI, and RAG systems. Skilled in building end to end ML pipelines, hybrid classification systems, and LLM powered applications. Strong foundation in Data Structures & Algorithms (DSA), competitive programming, and problem solving, along with experience in classical ML models and vector databases. Seeking opportunities in AI driven and data-centric environments.

Education

IIIT Bhubaneswar

Bachelor of Technology in Electronics and Telecommunication Engineering

• CGPA: 8.11 / 10.0

Bhubaneswar, Odisha

Aug 2023 – Jul 2027

Projects

Multi-Agent AI Research System | *LangChain, LCEL, Groq API, BeautifulSoup, Streamlit* | [GitHub](#) | [Live Demo](#)

- Implemented a **Search Agent** powered by **Tavily AI** and a **Reader/Scraper Agent** using **BeautifulSoup** to autonomously retrieve and extract relevant web content for research tasks.
- Designed a **Writer Chain** and **Critic Chain** using **LangChain LCEL Pipelines** and **Groq API** to generate structured research reports and automatically critique the final output for quality.
- Developed a **Streamlit** frontend and deployed the full application, demonstrating strong skills in **AI agent orchestration, API integration, deployment workflows, and environment variable security**.

RAG-Based Document Q&A System | *Python, LangChain, OpenAI API, Streamlit, HuggingFace* | [GitHub](#)

- Built a complete **Retrieval-Augmented Generation (RAG) system** from scratch capable of reading documents and answering questions intelligently using real-world production style architecture.
- Implemented an **end to end document ingestion pipeline** with chunking, embedding generation using **HuggingFace** models, and vector storage via **FAISS** for efficient semantic retrieval.
- Integrated **LLM based answer generation** using the OpenAI API with retrieved context, enabling accurate and grounded responses that go beyond the model's base knowledge.

Hybrid Log Classification System (Regex + ML + LLM) | *Python, NLP, BERT, FastAPI, Streamlit* | [GitHub](#)

- Designed and implemented a **hybrid log classification system** using Regular Expressions, **BERT**, and **LLMs** to categorize system logs into security alerts, resource usage, and workflow errors.
- Built a **multi stage pipeline** to dynamically route logs based on complexity, improving classification efficiency and reducing unnecessary model computation.
- Integrated **BERT based classification** and LLM (DeepSeek/Qwen) for handling complex and low data scenarios.
- Created a **Streamlit** frontend and **FastAPI** backend for real time log analysis and user interaction.

Technical Skills

Programming Languages: Python, C++, C, Java, SQL

AI/ML & GenAI: Machine Learning, Deep Learning, NLP, BERT, LLMs, RAG, LangChain, Prompt Engineering

Frameworks & Libraries: TensorFlow, Keras, Scikit-learn, Pandas, NumPy, Matplotlib, HuggingFace, FastAPI, Streamlit

Databases & Tools: FAISS, ChromaDB, Git, GitHub

Core Concepts: Data Structures & Algorithms, OOPs, Operating System, DBMS

Achievements & Certifications

- LeetCode Knight** with **300+** problems solved
- Rated **Codeforces Specialist**
- Data Analytics**–by Deloitte (April 2026) [View Certificate](#)
- Generative-AI**–by IBM [View Certificate](#)
- Agentic AI development**–by IBM (May 2026) [View Certificate](#)
- Introduction to Large Language Models**–by Google [View Certificate](#)

Leadership & Extracurricular

Placement Coordinator | *IIIT Bhubaneswar*

- Coordinating placement activities and mentoring peers for campus recruitment drives.